

Patient Name:	Ms. BASANTI DEVI 10078040	Sample ID:	60400127778
Age / Gender:	50 Years / Female	Lab ID:	N/A
Ref. Doctor:	N/A	Report Date:	N/A

PERFORMED.

CLINICAL INDICATION & SAMPLE DESCRIPTION
Clinical Indication: Not Specified
Sample Description: Whole Blood EDTA

KEY GENETIC FINDINGS

Gene & Transcript	Location	Variant	Zygosity	Phenotype	Clinical Significance
Next Generation Sequencing (Massively Parallel Sequencing)	Next Generation Sequencing (Massively Parallel Sequencing)	Next Generation Sequencing (Massively Parallel Sequencing)		Next Generation Sequencing (Massively Parallel Sequencing)	
EDTA Blood	EDTA Blood	EDTA Blood		EDTA Blood	

Test Description:	Chronic myeloid leukemia (CML) is characterized by the presence of the Philadelphia chromosome, the product of t(9;22)(q34;q11) translocation. This translocation results in BCR-ABL1 fusion protein with constitutive ABL1 tyrosine kinase activity. The kinase inhibitor imatinib (STI571, Gleevec®) inhibits ABL1 kinase activity and is now the standard of care for early phase CML. Prolonged treatment with imatinib can lead to drug resistance, especially in patients with advanced disease. A large portion of resistant patients have acquired point mutations in the ABL1 kinase domain that renders the kinase resistant to the drug. Sites of point mutations in ABL1 associated with imatinib resistance span the entire kinase domain but often cluster in important hotspots.(1,2) Total RNA was extracted and nested reverse transcription PCR was performed to detect the BCR-ABL1 transcript and ABL1 kinase domain (KD) region was thereafter sequenced by Next Generation Sequencing for detection of mutant alleles.
Methodology:	Next Generation Sequencing (Massively Parallel Sequencing)
Sample:	EDTA Blood

Test results and Interpretation:	PATHOGENIC MUTATION DETECTED
Mutation Nomenclature:	L248V
Coverage:	>500x

Sr. No.	Mutations covered	Mutation Result	Sensitivity to Imatinib	Mean fold Increase in IC50 (Imatinib)2	Sr. No.	Mutations covered	Mutation Result	Sensitivity to Imatinib	Mean fold Increase in IC50 (Imatinib)2
1	T315I	Negative	R	50.8	19	E275K	Negative	NA	
2	E255V	Negative	R	44.3	20	F359C	Negative	NA	4.3
3	Y253H	Negative	R	36.3	21	E450K	Negative	NA	3.9
4	E255K	Negative	R	30.7	22	M351T	Negative	S	3.8
5	Y253F	Negative	R	25.2	23	M244V	Negative	I	3.8
6	E373G	Negative	NA	20.6	24	E355G	Negative	I	3.4
7	L248V	Positive	NA	19.9	25	V379I	Negative	I	3.1
8	G321E	Negative	NA	18.1	26	E355A	Negative	NA	3.0

9	E279K	Negative	NA	16.5	27	M388L	Negative	NA	2.7
10	G250E	Negative	I	11.7	28	T315A	Negative	S	2.4
11	D276G	Negative	NA	11.2	29	M237I	Negative	NA	2.4
12	F486S	Negative	NA	8.4	30	F311L	Negative	S	2.2
13	H396P	Negative	S	7.1	31	L387F	Negative	NA	2.0
14	H396R	Negative	I	6.7	32	V299L	Negative	S	1.7
15	F359V	Negative	I	5.8	33	F317V	Negative	S	1.4
16	F317L	Negative	I	4.8	34	G250A	Negative	NA	1.3
17	Q252H	Negative	I	4.8					
18	L387M	Negative	S	4.3					

Novel Mutation	Nomenclature	Computationally Predicted Effect	
-	-	-	
REFERENCES:			

INTERPRETATION & REMARKS
Ref Id 2 : # For specimens received from non NCGM locations, it is presumed that it belongs to the patient as identified on the labels of the container/Test Requisition Form and it has been verified as per GCLP (Good Clinical Lab Practices) by the referrer at the time of collection of the specimen. NCGM's responsibility is limited to the analytical part of the assay performed. Dr. Arpan K. Mehta MD Path, PDF Lab. Haemat., PDF Mol. Haemonc. Printed On :09-May-2026 14:13 BCR::ABL1 KINASE DOMAIN MUTATION ANALYSIS (IRMA) BY NEXT GENERATION SEQUENCING

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Pathologist

Senior Geneticist

Medical Director